



INVENTORY MANAGEMENT SYSTEM GUIDE

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JANUARY 6, 2024
TECHINNOVATORS INC

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Technical Documentation

1. System Architecture and Design:

1.1.1 Overview

The Technology Innovators' inventory management system is important for enhancing workflow efficiency within the company. It centralizes the oversight of equipment used in daily projects, significantly reduces the time spent locating items, and minimizes losses due to equipment mismanagement. With its user-friendly interface and comprehensive features, the system optimizes resource allocation while promoting accountability and reliability.

1.1.2 Components

The system is comprised of several key components:

1.2.1. Python Programming Language: Python is utilized to develop the user interface, allowing employees to interact with the system seamlessly. Through this interface, employees can view their usage reports and sign in to return equipment. The development environment for this application is PYCHARM, which facilitates efficient coding and debugging.

1.2.2. Oracle APEX: Oracle APEX serves as the host platform for both the database and the administrative application. This platform enables authorized administrators to view, update, and assign inventory. The application on Oracle APEX is designed with good security measures, ensuring that only authorized personnel have access to sensitive inventory data.

1.2.3. API (Application Programming Interface): APIs play a crucial role in enabling seamless communication between the Python-based user interface and the Oracle APEX administrative application. These APIs ensure that data is synchronized in real time, providing a cohesive user experience and accurate inventory tracking.

1.2.4. SQL (Structured Query Language): SQL is employed to query and manage the database tables. It handles various operations such as equipment allocation and updates, ensuring that all changes are accurately reflected in the database.

Additionally, SQL was used in the development of the APIs within Oracle APEX, facilitating efficient data manipulation and retrieval.

1.2.5. PL/SQL (Procedural Language/Structured Query Language): PL/SQL is used to handle procedural logic within the database. It is instrumental in managing potential errors that may arise during various system functions, thereby enhancing the system's robustness and reliability.

These components work together to create a comprehensive and efficient inventory management system, ensuring smooth operations and reliable data management.

1.1.3 Interactions

i) Administrators

Administrators interact with the database management system and the Inventory of Equipment Management System application on Oracle APEX to perform various tasks, including:

- **Equipment Authorization:** Approving and managing access to equipment.
- **Updates:** Performing updates to keep the inventory database current and accurate.
- **Additions:** Adding new equipment and detailed information to the system.

ii) Employees

Employees utilize the Python-based system to:

- **Log Equipment Returns:** Signing in to record the return of equipment after use.
- **View Equipment:** Accessing their usage reports and current inventory status to check equipment availability and history.

For more detailed instructions, please refer to the user manuals and guides.

1.2 API Documentation:

The system features two key APIs:

1.2.1 Get-Report API: This API retrieves data from the INV_USAGE_RECORDS table in Oracle Apex and is used to help generate a detailed report on the Python-based interface. It empowers employees to conveniently access their individual usage histories through the Python interface, facilitating a thorough review of their usage records.

1.2.2 Post-Return API: Designed for seamless functionality, the Post-Return API facilitates the effortless posting of equipment return dates into INV_USAGE_RECORDS. This updating process occurs when employees return the equipment, utilizing their booking ID and QR code for identification. The API incorporates bind variables, including r_date, :booking_id, and :empid, ensuring efficient data management.

Refer to Appendix B

1.2.3 Access Control:

Access to the system's backend, which includes the database management system, inventory management system, and APIs, is strictly controlled and restricted to authorized personnel via the Oracle APEX platform. This stringent measure ensures the confidentiality and security of the company's data.

Authorized personnel must have valid APEX credentials to access these critical components of the system. Their responsibilities include managing equipment authorization, updates, and additions to the inventory.

Employees, on the other hand, have limited access. They can only view their inventory usage records within the Python-based application. This restriction upholds stringent data integrity standards and ensures that sensitive information remains secure.

For more detailed guidelines, please refer to the access control policies in the user manuals and guides.

1.3. Database Schema and Data Dictionary:

1.3.1 Tables:

INV_EMPLOYEES: Stores employee details, including employee ID, name, surname, and department. (view table structure below)

INV_EMPLOYEES						
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries						
+ Add Column Modify Column Rename Column Drop Column Refresh More						
Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
EMPLOYEE_ID	NUMBER	N	"WKSP_JFS354"."ISEQ\$\$_Z74593120".nextval	1		BY DEFAULT
EMPLOYEE_NAME	VARCHAR2(100 CHAR)	Y				
EMPLOYEE_SURNAME	VARCHAR2(100 CHAR)	Y				
DEPARTMENT	VARCHAR2(50 CHAR)	Y				

INV_EQUIPMENT_: Manages equipment inventory, tracking equipment ID, name, and quantity. (view table structure below)

INV_EQUIPMENT_						
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries						
+ Add Column Modify Column Rename Column Drop Column Refresh More						
Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
EQUIPMENT_ID	NUMBER	N	"WKSP_JFS354"."ISEQ\$\$_Z7641026".nextval	1		ALWAYS
EQUIPMENT_NAME	VARCHAR2(255 BYTE)	N				
EQUIPMENT_QUANTITY	NUMBER	N				

INV_USAGE_RECORDS: Records usage history, including booking ID, assignment and return dates, and foreign keys linking to employee and equipment IDs. (view table structure below)

INV_USAGE_RECORDS						
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries						
+ Add Column Modify Column Rename Column Drop Column Refresh More						
Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
BOOKING_ID	NUMBER	N	"WKSP_JFS354"."ISEQ\$\$_Z76415483".nextval	1		BY DEFAULT
DATE_BOOKED	DATE	N				
DATE_RETURNED	DATE	Y				
EMPLOYEE_ID	NUMBER	N				
EQUIPMENT_ID	VARCHAR2(10 CHAR)	N				

1.3.2 Data Dictionary:

API	The API enables communication between Apex and Python	APIs in the systems: - Get report.
-----	---	---

	environments, facilitating data exchange and function invocation.	- Post return.
Bind Variables	Placeholder variables in the database store data received from Python applications, such as booking ID and return date. They facilitate seamless data insertion into the database using PL/SQL.	:r_date :booking_id :empid
Python Library	Python libraries were employed to implement and execute the system's code effectively.	http.cient, opencv, pyzbar, datetime, time, tkinter, JSON, ttk message box.
Module	Modules in Apex serve as containers for organizing APIs. They act as repositories for APIs, providing a structured approach to access and manage them within the application.	Apex.Python

1.4 Third-Party Libraries and Dependencies:

1.4.1 Dependencies

- **API Integration:** The system leverages APIs to facilitate seamless communication between the database and the Python interface, ensuring efficient data exchange.

- **Python Libraries:** Various Python libraries are employed to develop the user interface and establish connectivity with Oracle APEX and the database.

1.4.2 Installation

System Versions:

This system has been developed using Oracle APEX version 23.2.4 and Python version 3.2 (64-bit). If implementing the system on different platforms or versions, adjustments may be necessary to account for potential variations in functionality.

Compatibility:

Ensure compatibility of all libraries and dependencies when installing the system on alternative environments to maintain optimal performance.

For detailed installation instructions and troubleshooting, please refer to the installation manuals and technical guides.

1.4.3 Python Libraries and Modules Utilized:

The Inventory Management System uses these Python tools:

- i) **http.client:** Helps send HTTP requests to communicate with the Oracle Apex API.
- ii) **cv2:** Captures images from the webcam and decodes QR codes.
- iii) **pyzbar:** Specifically decodes QR codes.
- iv) **datetime:** Gets the current date from the system.
- v) **time:** Adds delays when detecting QR codes.
- vi) **tkinter:** Essential for building the user-friendly interface (GUI) for employees.
- vii) **message box:** Shows error and success messages in the tkinter app.

2. Setup and Configuration Guides:

2.1 Hardware and Software Prerequisites

To make setting up and configuring the Inventory Management System easy, ensure you meet the following hardware and software requirements:

2.2.1 Hardware Prerequisites:

- A PC or Desktop capable of installing necessary software.
- Sufficient processing power and memory to handle system operations.
- Stable internet connection for efficient software operation.
- Webcam for barcode scanning.

2.2.2 Software Prerequisites:

- Python, PyCharm, Oracle Apex.
- Postman website for verifying API URLs.
- Required Python libraries.
- Web browser for using Oracle Apex.

2.3 Setup Instructions:

Follow these steps to set up the Inventory Management System:

2.3.1 Download and Install Python and PyCharm:

- Download Python from python.org.
- Download PyCharm from jetbrains.com/python.
- Install Python by locating and running the installer file.
- Follow the installation wizard instructions.

2.3.2 Download Required Python Libraries:

- Use the command line to install libraries: 'pip install {library name}'.

2.3.3 Install PyCharm:

- Download and Install the PyCharm Community for your operating system.
- Complete the installation wizard prompts.

2.3.4 Sign Up on APEX:

- Sign up at apex.oracle.com.
- Create a workspace and refer to online resources for assistance.

2.3.5 Setting up Python:

- Create a New Project in PyCharm.
- Create two **.py** Files: One for managing inventory usage history and the other for updating return dates.

Refer to the appendix for code examples.

2.4 API Configuration: Step-by-Step Guide

2.4.1 Get Report API

1. Log in to Oracle APEX:

- Go to [Oracle APEX] (<https://apex.oracle.com/>).
- Sign in with your credentials.

2. Navigate to SQL Workshop:

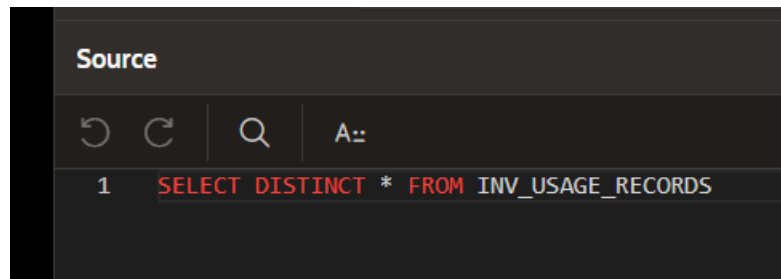
- Click on SQL Workshop.
- Select RESTful Services.

3. Create Module:

- Select Modules > Create Module.
- Enter module details:
 - Name: `InventoryManagementAPI`
 - Base Path: `/api/inventory`
- Click Create Module.

4. Create URL Template:

- Click on your module.
- Click Create Template.
- Enter URI Template: (e.g., `/getReport`)
- Click Create Template.
- Click Create Handler.
- Set Method to `GET`.
- Insert your SQL query as the Source:



- Click Create Handler.

5. Test in Postman:

- Copy the generated handler URL.
- Go to [Postman](<https://web.postman.co>).
- Navigate to Workspace > My Workspace.
- Create a New Collection.
- Create a GET request with the copied URL.
- Check for a success code `200`.

Post Return API:

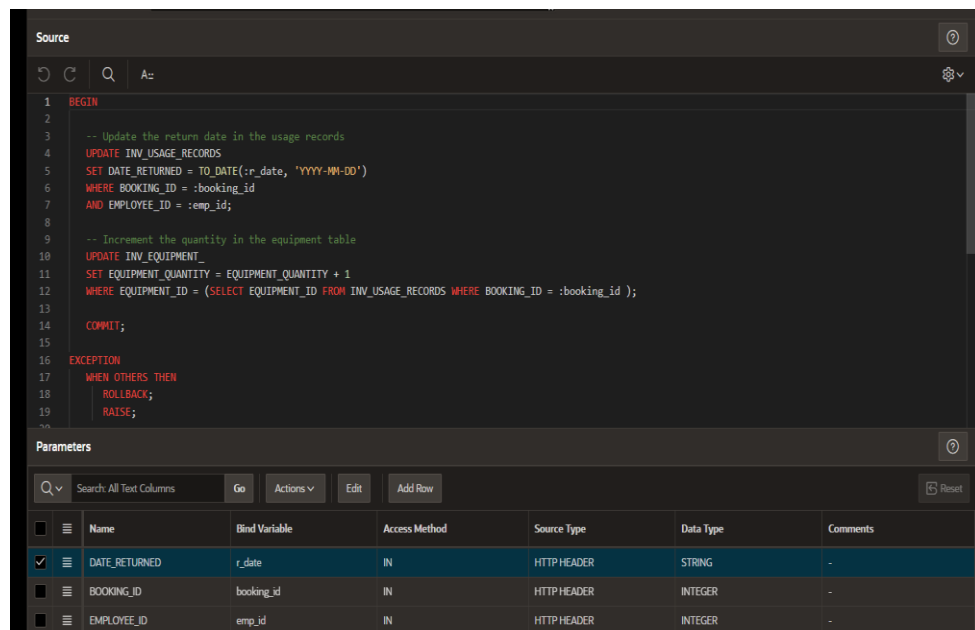
1. Create URL Template:

- In the same module, click Create Resource Template.
- Enter URI Template: (e.g., `/postReturn`)
- Click Create Template.

2. Create Handler:

- Click Create Handler.

- Set Method to `POST`.
- Insert your PL/SQL query as Source and bind variables as the Parameters:



- Click Create Handler.

3. Test in Postman:

- Copy the generated handler URL.
- Go to [Postman](<https://web.postman.co>).
- Navigate to Workspace > My Workspace.
- Create a New Collection.
- Create a POST request with the copied URL.
- Insert bind variables in the body (e.g., `r\_date`, `booking\_id`, `empid`).
- Check for a success code `200`.

By following these concise steps, you can configure your APIs in Oracle APEX for secure and efficient database communication.

2.5 Configuration Settings and Options:

- **API Integration:** Configure endpoints and authentication for database communication.
- **User Permissions:** Define access levels for data security.

- **Webcam Settings:** Adjust webcam settings for better QR code scanning.
- **Database Configuration:** Optimize settings for performance.

By following these steps, you can efficiently deploy the Inventory Management System. Regularly review and update configurations to meet changing needs and security concerns.

3. User Manuals and Guides

3.1 End-User Documentation:

The inventory management system provides the following features and usage guidelines for end users:

Features:

- View Inventory Usage History
- Return Inventory

Guidelines:

- To access their inventory usage history report, users (employees) simply provide their employee ID.
- To request equipment assignment, users need to request authorization from a company administrator. Upon approval, users receive a booking ID to note down.
- To return equipment, users provide their booking ID received during equipment assignment and scan a unique QR code containing their employee ID.
- For generating QR codes, qr-code-generator.com is recommended. Ensure to insert your employee ID before generating and downloading the code on your mobile device.

Note: If there's no record of equipment return, the administrator assumes possession. After the stipulated return period, company policy on equipment usage applies.

How the Python-based Application Works:

View Inventory Usage History:

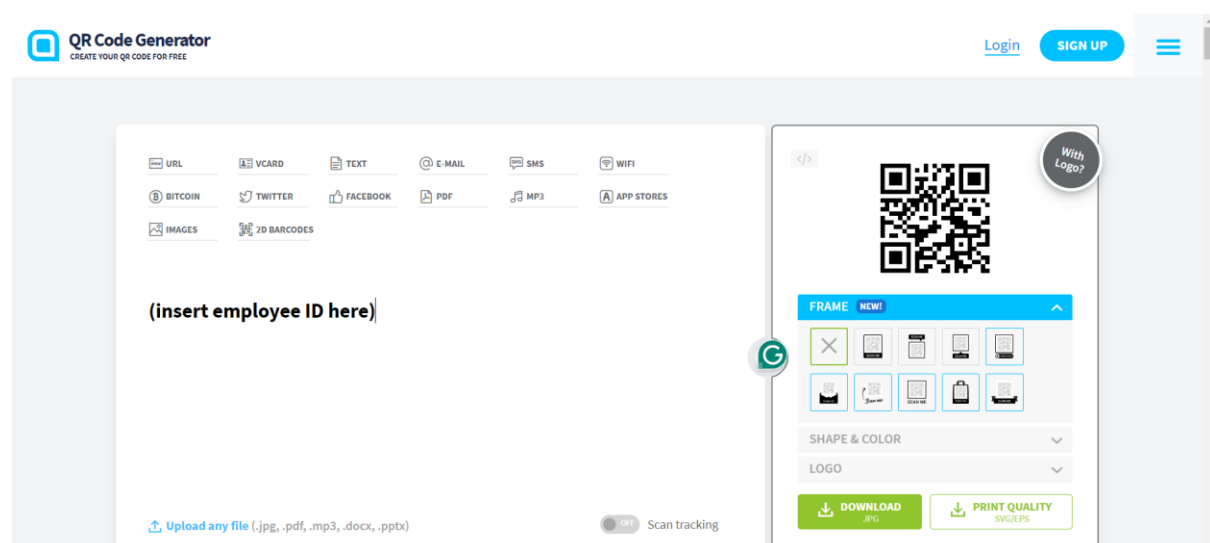
1. Run the code.
2. Input a valid employee ID when prompted.
3. The system displays all equipment usage records.
4. Code execution ends.

Return Equipment:

1. Run the code.
2. Enter the booking ID when prompted.
3. After a few seconds, the webcam opens to scan the unique QR code with the employee ID.
4. The system registers the equipment return and updates the return date column.

Generate QR Code:

1. Visit qr-code-generator.com.
2. Input your employee ID and generate the QR code.
3. It's best to do this on your mobile device.



3.2 Administrator Guides:

In addition to the end-user documentation, administrators have access to advanced features for managing the system, including:

- Authorization of equipment allocation requests.
- Addition and updating of equipment inventory.
- Viewing inventory usage history for all employees.
- Viewing all equipment within the company.

Guidelines:

To access the Tech Inn Inventory Management System Application, administrators need to log into the Oracle Apex environment with their unique login details.

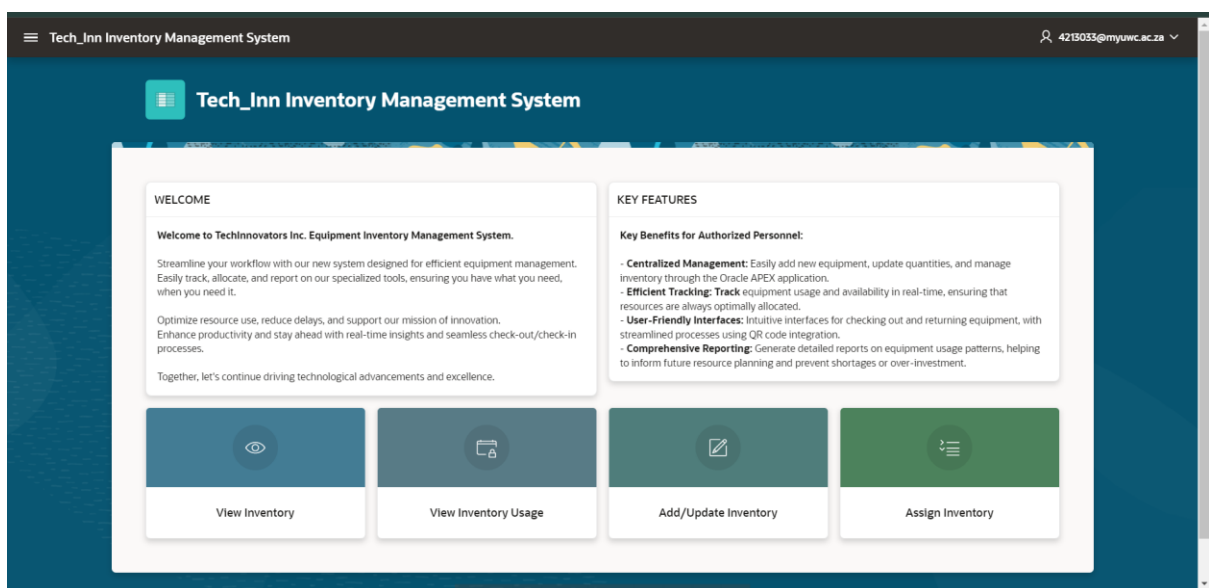
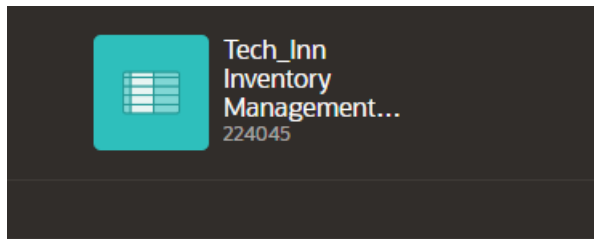
How to Use the Apex Application:

1. Navigate through the application using the navigation menu.
2. View records from the view pages.
3. Add Equipment: Input equipment name and quantity. The ID generates automatically upon insertion of a new entry.
4. Update Inventory: Modify equipment quantities as needed.
5. Assign Equipment: Allocate equipment to employees by specifying the equipment quantity and entering the employee ID.
6. Provide the booking ID to employees for future reference.

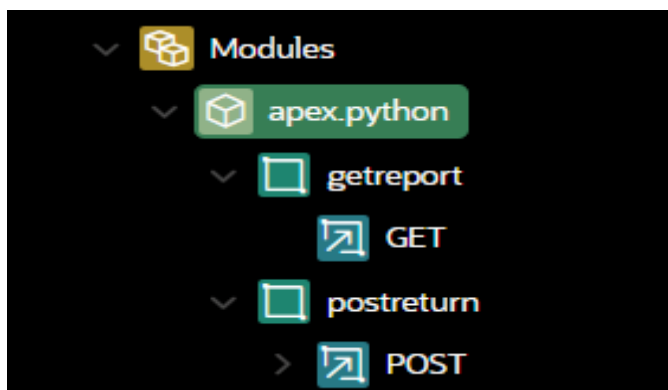
Appendices

A)

Application Interface



B)



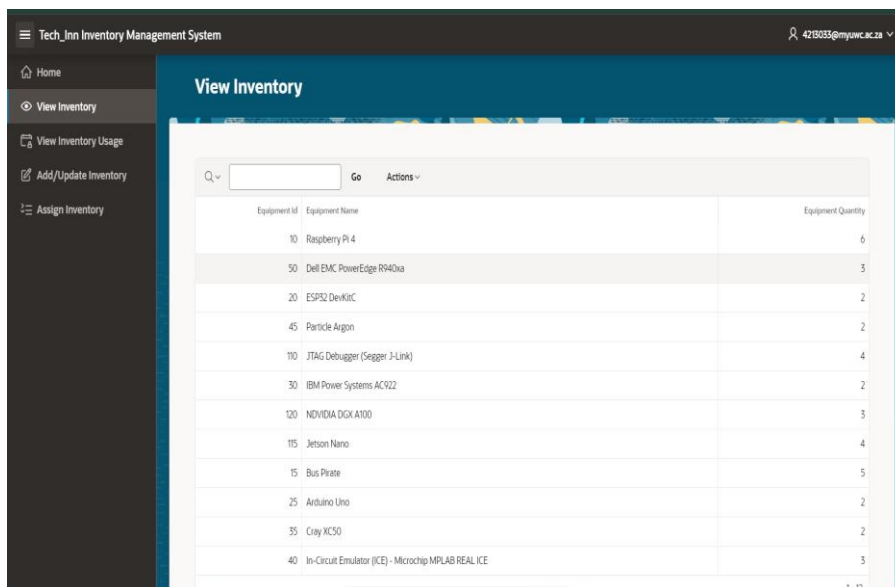
The above shows the module that contains the two APIs within the Python environment.

C)

View Report code

```
SQL Query

SELECT
    EQUIPMENT_ID,
    EQUIPMENT_NAME,
    EQUIPMENT_QUANTITY
FROM
    INV_EQUIPMENT_;
```

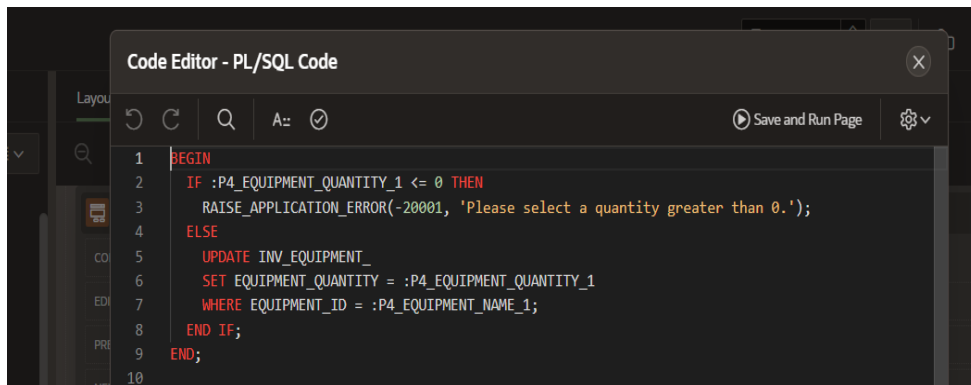


Equipment Id	Equipment Name	Equipment Quantity
10	Raspberry Pi 4	6
50	Dell EMC PowerEdge R940xa	3
20	ESP32 DevKitC	2
45	Particle Argon	2
110	JTAG Debugger (Segger J-Link)	4
30	IBM Power Systems AC922	2
120	NVIDIA DGX A100	3
115	Jetson Nano	4
15	Bus Pirate	5
25	Arduino Uno	2
35	Cray XC50	2
40	In-Circuit Emulator (ICE) - Microchip MPLAB REAL ICE	3

D)

Update Equipment

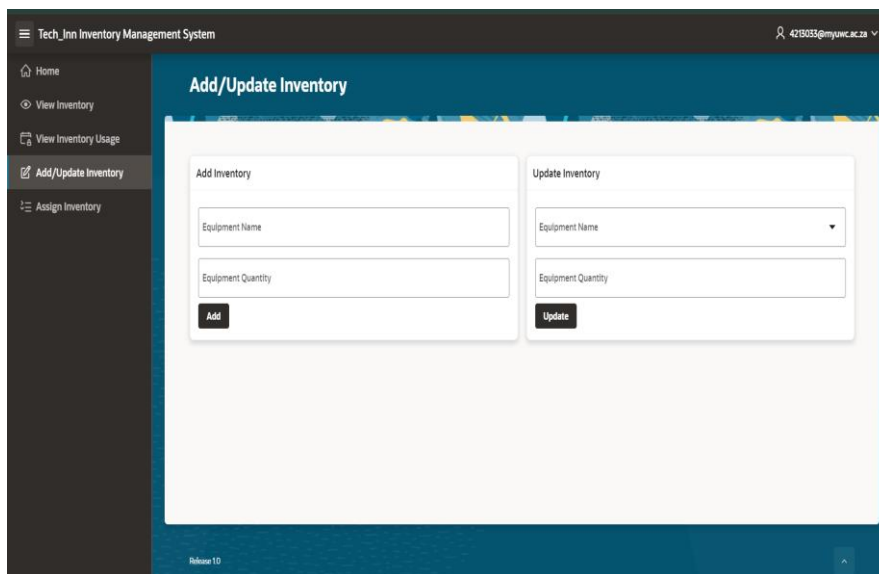
i) Back-end code for updating equipment with error handling.



```
1 BEGIN
2 IF :P4_EQUIPMENT_QUANTITY_1 <= 0 THEN
3 RAISE_APPLICATION_ERROR(-20001, 'Please select a quantity greater than 0.');
```

```
4 ELSE
5 UPDATE INV_EQUIPMENT
6 SET EQUIPMENT_QUANTITY = :P4_EQUIPMENT_QUANTITY_1
7 WHERE EQUIPMENT_ID = :P4_EQUIPMENT_NAME_1;
8 END IF;
9 END;
```

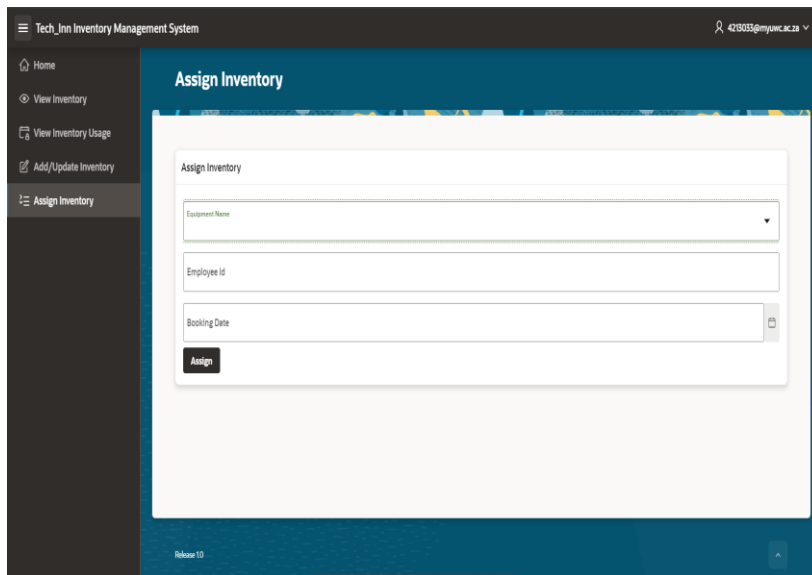
ii) User Interface



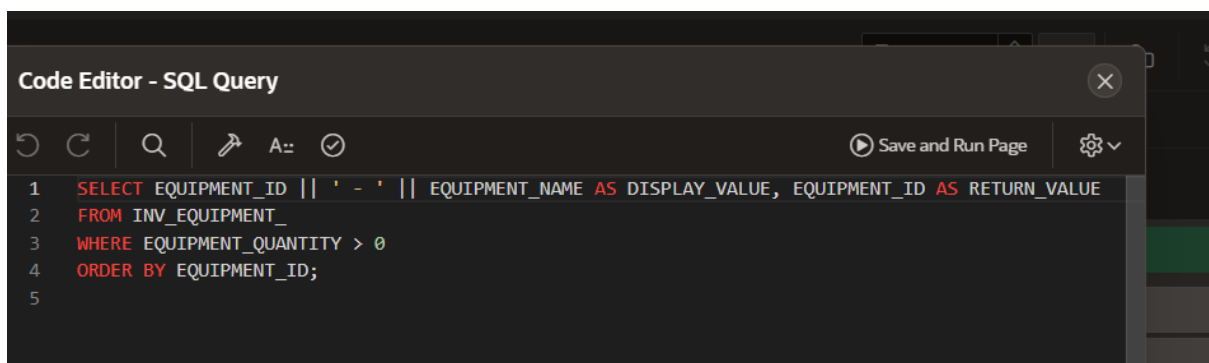
E)

Assign Equipment interface. The code for the select list ensures that only equipment with a quantity greater than zero is shown. If the required equipment is not on this list, it is not available for assignment.

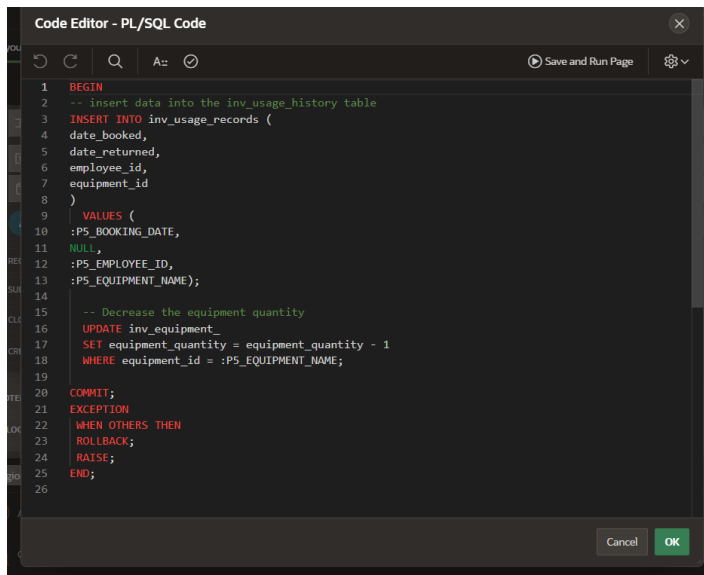
i)



ii)



iii)



```
1 BEGIN
2 -- insert data into the inv_usage_history table
3 INSERT INTO inv_usage_records (
4   date_booked,
5   date_returned,
6   employee_id,
7   equipment_id
8 )
9   VALUES (
10    :P5_BOOKING_DATE,
11    NULL,
12    :P5_EMPLOYEE_ID,
13    :P5_EQUIPMENT_NAME);
14
15 -- Decrease the equipment quantity
16 UPDATE inv_equipment
17 SET equipment_quantity = equipment_quantity - 1
18 WHERE equipment_id = :P5_EQUIPMENT_NAME;
19
20 COMMIT;
21 EXCEPTION
22 WHEN OTHERS THEN
23   ROLLBACK;
24   RAISE;
25 END;
26
```

F)

